

PART H: FEATURE CONSTRUCTION & TRANSFORMATION

Final dataset save:

```
Dataset Saved Successfully
```

Dataset shape check, Columns check, Missing value final check:

```
Dataset Saved Successfully
['customer_id', 'age', 'gender', 'region', 'education_level', 'employment_type', 'annual_income', 'loan_amount', 'loan_purpose', 'credit_score', 'repayment_history', 'transaction_count', 'spending_ratio', 'join_date', 'default_flag', 'income_missing', 'Year', 'Month', 'Day', 'Weekday', 'education_level_encoded', 'gender_encoded', 'income_bin', 'income_binary', 'income_quantile', 'income_cluster', 'income_standard', 'income_minmax', 'income_maxabs', 'income_robust', 'income_log', 'income_reciprocal', 'income_sqrt', 'income_power', 'debt_to_income', 'avg_monthly_transactions', 'spending_to_income']
customer_id      0
age              0
gender           0
region           0
education_level  0
employment_type  0
annual_income    0
loan_amount      0
loan_purpose       0
credit_score     0
repayment_history 0
transaction_count 0
spending_ratio   0
join_date        0
default_flag     0
income_missing   0
Year             0
Month            0
Day              0
Weekday          0
education_level_encoded 0
gender_encoded   0
income_bin       0
income_binary    0
income_quantile  0
income_cluster   0
income_standard  0
income_minmax    0
income_maxabs    0
income_robust    0
income_log       0
income_reciprocal 0
income_sqrt      0
income_power     0
debt_to_income   0
avg_monthly_transactions 0
spending_to_income 0
dtype: int64
Final Dataset Shape : (1000, 37)
```